

EE210 Analog Electronics: Project 2

Constant gm Bias(5T-OTA 10ms Target)

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Abstract—This paper presents the design of a fully differential 5-transistor operational transconductance amplifier (OTA) implemented in a 180 nm CMOS process. The primary objective is to achieve a transconductance of 10 mS with less than 1% variation over a temperature range of -20°C to 100°C . To address the temperature dependence of device parameters, a constant- g_m biasing scheme based on a beta-multiplier is used. The design was developed iteratively, starting from a basic OTA and progressively integrating the bias network and common-mode feedback.

To further improve temperature stability, the beta-multiplier ratio was increased, enabling partial cancellation of higher-order temperature effects. A continuous-time differential difference amplifier (DDA)-based common-mode feedback (CMFB) circuit is used to regulate the output common-mode voltage without introducing resistive loading at the output nodes.

Simulation results show that the final design meets the target g_m with less than 1% variation across temperature, while maintaining stable operation and high output impedance.

I. INTRODUCTION

Operational transconductance amplifiers (OTAs) are key building blocks in continuous-time analog systems, with applications ranging from filters and data converters to sensor interfaces. In such systems, parameters like bandwidth and linearity are directly influenced by the transconductance (g_m) of the OTA. This work focuses on the design of a fully differential 5-transistor OTA in a standard 180 nm CMOS process, targeting a transconductance of 10 mS.

While achieving a high g_m at nominal conditions is relatively straightforward, maintaining it across temperature is significantly more challenging. As temperature increases from -20°C to 100°C , carrier mobility (μ) degrades, leading to a noticeable reduction in g_m for a conventionally biased OTA. In addition, the use of a fully differential architecture introduces the need for accurate control of the output common-mode voltage. Without a proper feedback mechanism, the output nodes (being high impedance) can drift, potentially driving transistors out of saturation and degrading performance.

To address these issues, the design was developed through a structured iterative approach. The initial phase established a baseline using a diode-connected 5T OTA to validate the target transconductance at room temperature. This was followed by the design of a constant- g_m biasing block based on a beta-multiplier, aimed at reducing the temperature dependence of the bias current. The ratio within this block was later increased to improve compensation of higher-order temperature effects. In the final stages, the OTA was integrated with active loads

and a continuous-time differential difference amplifier (DDA)-based common-mode feedback (CMFB) circuit, enabling stable operation while maintaining high output impedance.

II. PHASE 1: BASELINE 5T OTA DESIGN AND CHARACTERIZATION

The objective of this phase was to establish a functional baseline for the OTA architecture. A standard 5-transistor (5T) OTA with diode-connected PMOS loads was used to verify that the target transconductance of 10 mS could be achieved at nominal conditions, before introducing temperature compensation or common-mode feedback.

A. Operating Region and Initial Current Estimation

Achieving a transconductance of 10 mS requires careful selection of the operating region to avoid excessive power consumption or impractically large device sizes. Instead of selecting bias currents arbitrarily, the design was guided using the transconductance efficiency (g_m/I_d) methodology.

From the g_m/I_d versus V_{GS} characteristics, the moderate inversion region was identified as a suitable operating point, offering a balance between current efficiency and speed. An efficiency target of approximately 17 V^{-1} was selected.

With this choice, the required branch current can be estimated as:

$$I_d = \frac{g_m}{(g_m/I_d)} \quad (1)$$

This provides an initial estimate for the bias current, which is later refined through simulation. Based on this approach, the design converged to a branch current of approximately $606\text{ }\mu\text{A}$, corresponding to a total tail current near 1.2 mA.

B. Biasing Strategy and Voltage Allocation

To ensure that all transistors remain in saturation while preserving voltage headroom, the DC bias voltages were defined prior to final sizing:

- **Input Common-Mode Bias:** The input gates were biased at 900 mV ($V_{dd}/2$), centering the input range and allowing for symmetric signal swing.
- **Tail Bias Voltage (V_{tail}):** The gate of the tail transistor was biased at 600 mV. Based on I_d - V_{GS} characteristics, this ensures sufficient overdrive to support the required tail current while maintaining saturation.
- **Voltage Headroom:** With the input gates at 900 mV and the input devices operating in moderate inversion,

the common-source node settles at a lower potential, providing adequate V_{DS} for the tail device to remain in saturation.

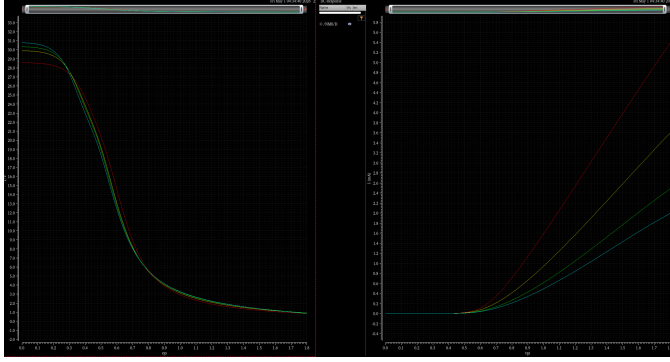


Fig. 1. g_m/I_d versus V_{GS} characteristics for NMOS devices, used to identify the moderate inversion region and select the operating efficiency for the input differential pair.

C. Device Sizing Methodology

With the operating region, current levels, and bias voltages defined, device dimensions were first estimated analytically and then refined through simulation.

1) *Input Differential Pair*: To reduce channel-length modulation and improve intrinsic gain ($g_m r_o$), the channel length of the input pair was chosen as 540 nm ($3 \times L_{min}$). The device width was estimated using the square-law approximation:

$$g_m = \mu_n C_{ox} \frac{W}{L} V_{ov} \quad (2)$$

Rearranging,

$$W = \frac{g_m \cdot L}{\mu_n C_{ox} \cdot V_{ov}} \quad (3)$$

At the selected efficiency, the corresponding overdrive voltage is relatively small, which leads to a large required device width. In practice, this width was implemented using a multi-finger layout ($m = 10$) to reduce gate resistance and improve layout feasibility.

2) *Tail Current Source*: The tail transistor was assigned the same channel length (540 nm) to improve its output resistance. A higher output resistance helps maintain a stable bias current and improves common-mode rejection.

3) *Diode-Connected Loads and Output Centering*: Diode-connected PMOS devices were used as loads to provide simple self-biasing. Their channel length was kept at the minimum (180 nm) to limit parasitic capacitance at the output nodes.

The devices were sized to establish a suitable V_{sg} drop, which in turn sets the output common-mode level. This ensures that the input NMOS pair maintains sufficient V_{DS} to remain in saturation during operation.

D. Baseline Results and Thermal Limitations

DC operating point simulations confirmed that the OTA achieves a transconductance of approximately 10 mS at room temperature, validating the initial design.

However, key limitations were observed:

- **Temperature Sensitivity**: A temperature sweep revealed a strong dependence of g_m on carrier mobility (μ). As temperature increases, mobility decreases, resulting in a noticeable reduction in transconductance.

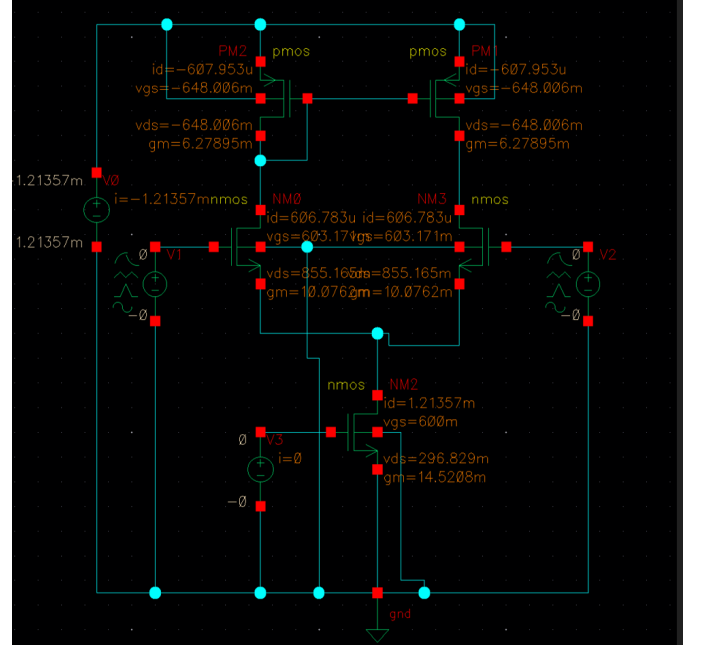


Fig. 2. Baseline 5T OTA schematic used in Phase 1.

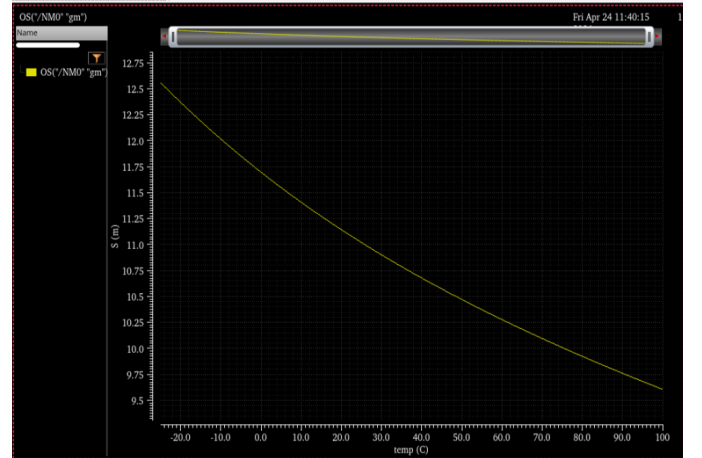


Fig. 3. Transconductance variation with temperature in Phase 1, showing significant drift across the operating range.

The observed variation in g_m across the -20°C to 100°C range does not meet the $< 1\%$ stability requirement. This motivates the transition to Phase 2, where a constant- g_m biasing scheme is introduced, along with improved load structures to enhance gain.

III. PHASE 2: CONSTANT- g_m BIASING AND IDEAL CMFB INTEGRATION

Phase 1 established the baseline device sizing required to achieve a transconductance of 10 mS. However, the architecture exhibited strong temperature dependence due to carrier mobility degradation and limited DC gain due to the diode-connected loads.

To address these limitations, Phase 2 introduces a Constant- g_m (Beta-Multiplier) bias core and replaces the passive loads with active current sources. An ideal Common-Mode Feedback (CMFB) network is temporarily used to stabilize the output.

A. The Beta-Multiplier: Addressing Thermal Drift

As temperature increases, carrier mobility (μ) decreases, which directly reduces the transconductance in a constant-current biased OTA. To compensate for this, a Beta-Multiplier bias core was implemented.

Instead of enforcing a fixed current, the Beta-Multiplier forms a self-biased feedback loop that ties the transconductance of the devices to a reference resistor (R_s). Since resistors are significantly more stable over temperature compared to transistor parameters, this reduces the sensitivity of g_m to temperature variations.

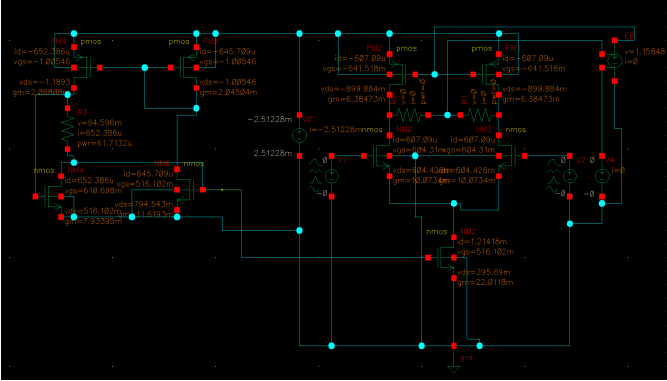


Fig. 4. Phase 2 integrated architecture with Beta-Multiplier biasing, active loads, and ideal CMFB stabilization.

B. Loop Operation and Derivation

To understand its behavior before full integration, the Beta-Multiplier was first analyzed independently. The loop consists of a symmetric PMOS current mirror and an asymmetric NMOS pair.

A scaling factor of $K = 4$ was selected for the NMOS devices. Let M_{b1} have width W and M_{b2} have width $4W$, with a source degeneration resistor R_s .

Assuming saturation operation:

$$I_{Mb1} = k' \left(\frac{W}{L} \right) (V_{GS,b1} - V_T)^2 \quad (4)$$

$$I_{Mb2} = k' \left(\frac{4W}{L} \right) (V_{GS,b2} - V_T)^2 \quad (5)$$

The PMOS mirror forces:

$$I_{Mb1} = I_{Mb2} = I \quad (6)$$

Applying KVL:

$$V_{GS,b1} - V_{GS,b2} = IR_s \quad (7)$$

Solving the system:

$$V_{GS,b1} - V_T = 2(V_{GS,b2} - V_T) \quad (8)$$

Substituting:

$$\frac{2I}{V_{GS,b1} - V_T} = \frac{1}{R_s} \quad (9)$$

Using $g_m = \frac{2I}{V_{GS} - V_T}$:

$$g_m = \frac{1}{R_s} \quad (10)$$

This shows that the transconductance is set directly by the resistor, making it less sensitive to process and temperature variations.⁴

C. Common-Mode Feedback (CMFB) Requirement

To prevent uncontrolled drift of the output common-mode voltage, a Common-Mode Feedback (CMFB) loop is introduced.

The CMFB senses the average output voltage and adjusts the gate bias of the PMOS load devices to maintain a fixed operating point. This ensures that the currents in the top and bottom branches remain balanced, keeping all transistors in saturation.

In this phase, an ideal voltage-controlled voltage source (VCVS) is used to implement the CMFB. This allows the output common-mode voltage to be fixed without introducing additional circuit complexity, enabling focused evaluation of the Beta-Multiplier biasing.

D. Results and Discussion

The introduction of the Beta-Multiplier biasing scheme results in a significant improvement in transconductance stability across temperature.

Unlike the baseline design in Phase 1, where g_m exhibited strong dependence on temperature due to carrier mobility degradation, the Phase 2 architecture maintains a nearly constant transconductance over the entire -20°C to 100°C range.

This behavior confirms that the bias current now adapts with temperature in a way that compensates for mobility variation, effectively stabilizing g_m . The residual variation is significantly reduced compared to the baseline and falls within a narrow band around the target 10 mS.

Overall, Phase 2 demonstrates that the Constant- g_m biasing approach successfully addresses the primary limitation observed in Phase 1, establishing a stable operating point across temperature while preserving the intended transconductance level. This behavior confirms that the bias current now adapts with temperature in a way that compensates for mobility variation, effectively stabilizing g_m . The residual variation is



Fig. 5. Transconductance(NM0) variation with temperature after introducing the Beta-Multiplier. The curve remains nearly flat, indicating strong suppression of thermal drift compared to Phase 1.

significantly reduced compared to the baseline and falls within a narrow band around the target 10 mS.

Overall, Phase 2 demonstrates that the Constant- g_m biasing approach successfully addresses the primary limitation observed in Phase 1, establishing a stable operating point across temperature while preserving the intended transconductance level.

IV. PHASE 3: OPTIMIZED BIAS SCALING AND SENSITIVITY ANALYSIS

While Phase 2 demonstrated that the Beta-Multiplier can stabilize g_m , this phase focuses on improving precision and reducing residual variation across temperature. The objective here is to refine the bias parameters such that the target transconductance of 10 mS is maintained with minimal sensitivity to component variations.

A. Sensitivity Analysis and Scaling of K

During iterative refinement, it was observed that increasing the reference resistance (R_s) improves the stability of g_m . This can be understood from the expression for transconductance in the Beta-Multiplier:

$$g_m = \frac{2}{R_s} \left(1 - \frac{1}{\sqrt{K}} \right) \quad (11)$$

Taking the derivative with respect to R_s :

$$\frac{dg_m}{dR_s} = -\frac{2}{R_s^2} \left(1 - \frac{1}{\sqrt{K}} \right) \quad (12)$$

This shows that the sensitivity of g_m to variations in R_s decreases with increasing resistance. In other words, for a

given absolute variation in R_s , the resulting variation in g_m becomes smaller as R_s increases.

However, increasing R_s alone reduces g_m . To maintain the target value of 10 mS while using a larger and more stable resistance, the scaling factor K was increased from 4 to 16.

For $K = 4$:

$$g_m = \frac{1}{R_s}$$

For $K = 16$:

$$g_m = \frac{2}{R_s} \left(1 - \frac{1}{4} \right) = \frac{1.5}{R_s}$$

This allows the use of a higher R_s while preserving the desired transconductance, effectively reducing sensitivity to resistor variations and improving overall stability.

B. Bias Distribution and OTA Integration

The updated architecture includes a bias distribution network that replicates the stabilized transconductance within the OTA branches.

An internal feedback loop within the bias circuit ensures that reference currents are well-defined and consistently mirrored. Currents generated in the Beta-Multiplier core are distributed and scaled through current mirrors to establish the required biasing conditions in the OTA.

The tail current of the OTA is derived from this network, ensuring that it tracks the same reference used to define g_m . Additionally, careful sizing and bias matching ensure that the operating conditions of the OTA input devices closely mirror those of the reference devices in the bias core.

This matching ensures that the transconductance of the OTA input pair follows the same relationship set by the Beta-Multiplier, effectively transferring the temperature-stable behavior into the main signal path.

C. Results and Performance Optimization

The introduction of the $K = 16$ scaling and the refined bias distribution network results in a clear improvement in transconductance stability.

Two observations are evident from the temperature sweep:

- The bias core devices still show a gradual variation in g_m with temperature, reflecting underlying device-level dependencies.
- The OTA input pair, however, maintains a nearly constant transconductance close to 10 mS, indicating effective compensation by the bias network.

This behavior confirms that the feedback and current distribution within the bias network suppress the impact of these variations before they propagate to the OTA.

Additionally, increasing R_s through the $K = 16$ scaling reduces sensitivity to resistor variations, further improving stability.

Overall, Phase 3 significantly reduces residual error and achieves a near temperature-independent transconductance across the specified range.

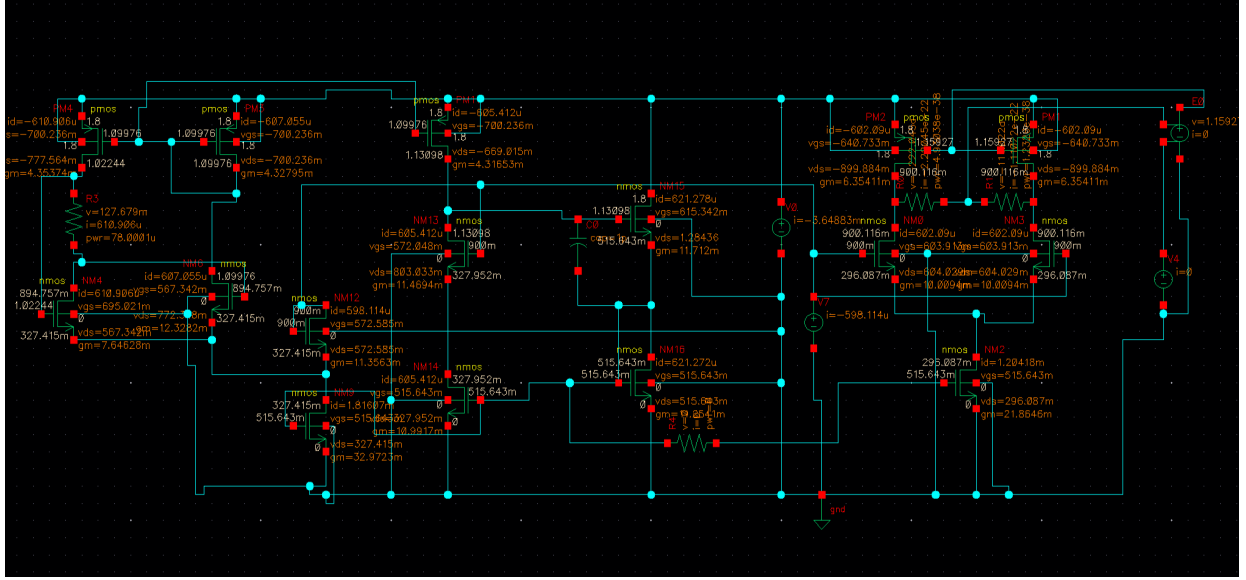


Fig. 6. Complete Phase 3 circuit implementation, including the Beta-Multiplier bias core and bias distribution network integrated with the OTA.

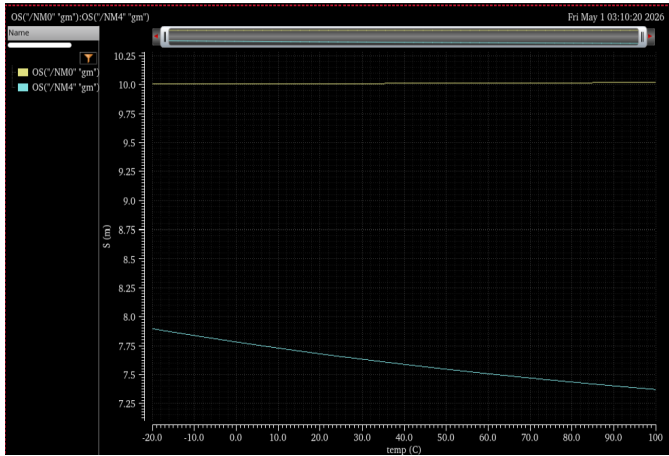


Fig. 7. Transconductance behavior in Phase 3. The OTA input(NM0) pair remains nearly constant, while the bias core(NM4) devices exhibit residual temperature dependence.

V. PHASE 4: FULL SYSTEM INTEGRATION AND TRANSISTOR-LEVEL CMFB

Phase 4 represents the final implementation of the OTA system. In this stage, the ideal VCVS used in earlier phases is replaced with a fully transistor-level Common-Mode Feedback (CMFB) circuit, resulting in a complete and physically realizable design.

A. DDA-Based CMFB Architecture

The implemented CMFB follows a Differential Difference Amplifier (DDA) architecture, consisting of three main functional stages:

- **High-Impedance Sensing:** A differential pair (Mc_1 – Mc_2) senses the output voltages (V_{op} , V_{on}) directly at their gates. This provides a high input

impedance, allowing the common-mode level to be monitored without loading the output nodes.

- **Comparison and Amplification:** The sensed common-mode voltage is compared with a reference voltage (V_{ref}). To improve tracking, the reference is passed through a replica PMOS source follower to match the DC behavior of the sensing path. The resulting error signal is amplified using the CMFB amplifier stage (Mc_3 , Mc_4 , Mc_{11} , Mc_{12}).
- **Frequency Compensation and Feedback:** A compensation network, formed using a MOS capacitor (C_B) and a PMOS device operating in the triode region (Mc_9), stabilizes the loop. The generated control voltage V_{cmfb} adjusts the gate bias of the PMOS load devices, regulating the output common-mode voltage.

B. System Integration and Design Perspective

The final architecture follows a structured division of roles:

- **Constant- g_m Bias Core:** Sets the transconductance through the $K = 16$ scaled Beta-Multiplier, providing a temperature-stable reference.
- **OTA Core:** Uses the bias current from the core to realize the desired transconductance with high output resistance.
- **CMFB Loop:** Maintains the output common-mode voltage without introducing resistive loading, preserving the OTA gain.

C. Results: Comparison with Ideal Implementation

The performance of the fully integrated system closely matches that of the ideal CMFB implementation used in Phase 3.

- **Thermal Stability:** The OTA input pair maintains a nearly constant transconductance close to 10 mS across the -20°C to 100°C range.

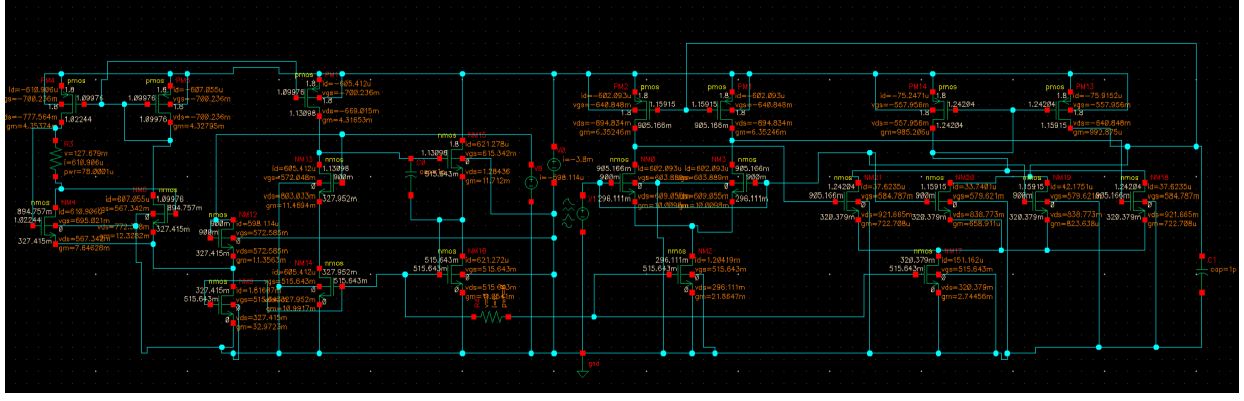


Fig. 8. Fully integrated OTA system including the Beta-Multiplier bias core, OTA, and transistor-level DDA-based CMFB.

- **Bias Compensation:** While devices within the bias core continue to exhibit temperature-dependent behavior, the OTA transconductance remains stable due to the feedback and biasing scheme.
- **Common-Mode Regulation:** The CMFB loop maintains the output common-mode voltage with minimal deviation, ensuring stable operation without degrading gain.

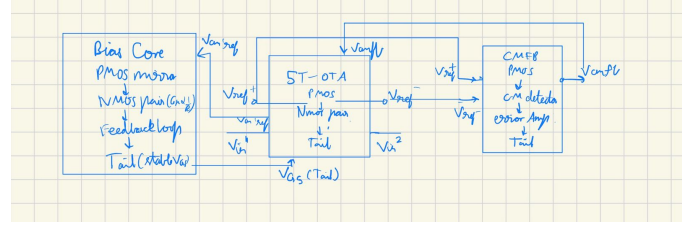


Fig. 10. System-level block diagram of the proposed OTA architecture, showing the constant- g_m bias core, ST OTA, and CMFB loop. The bias core generates a temperature-stable reference, which is mirrored into the OTA, while the CMFB loop regulates the output common-mode voltage.

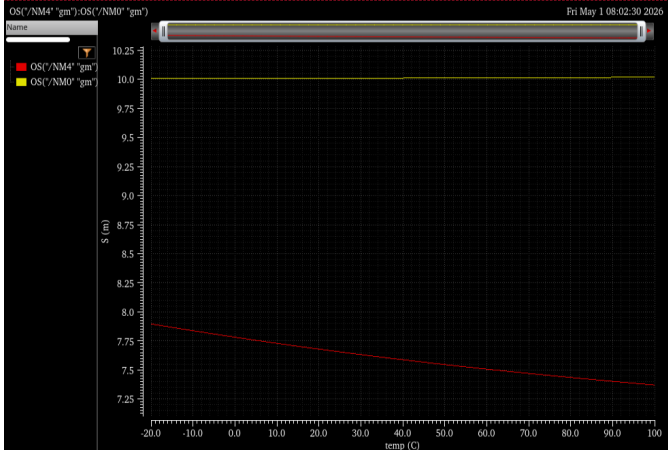


Fig. 9. Final transconductance variation with temperature, showing consistent behavior with the transistor-level CMFB implementation.

D. Conclusion

The final design meets the target transconductance of 10 mS with less than 1 percent error while maintaining strong stability across temperature. The combination of the $K = 16$ constant- g_m biasing scheme and the non-loading DDA-based CMFB results in a robust OTA suitable for practical implementation.

E. Architecture and Operating Point Summary

TABLE I
OPERATING POINT SUMMARY OF OTA CORE DEVICES

Device	I_{DS} (μA)	V_{GS} (mV)	V_{DS} (mV)	g_m (mS)	V_{th} (mV)	Reg
NM0 (In)	602.09	603.89	609.06	10.01	≈ 420	S
NM3 (In)	602.09	603.89	609.06	10.01	≈ 420	S
PM1 (Ld)	-602.09	-640.85	-894.83	6.35	≈ -450	S
PM2 (Ld)	-602.09	-640.85	-894.83	6.35	≈ -450	S
NM2 (Tail)	1204.19	515.64	296.11	21.86	≈ 412	S

TABLE II
OPERATING POINT SUMMARY OF BIAS NETWORK DEVICES

Device	I_{DS} (μA)	V_{GS} (mV)	V_{DS} (mV)	g_m (mS)	V_{th} (mV)	Reg
NM4 (Mstr)	610.91	695.02	567.34	7.65	≈ 421	S
PM4 (Mir)	-610.91	-700.24	-777.56	4.35	≈ -465	S
PM3 (Mir)	-607.06	-700.24	-700.24	4.32	≈ -465	S
NM13 (Sp)	605.41	572.05	803.03	11.47	≈ 422	S
NM12 (Sp)	598.11	572.59	572.59	11.36	≈ 422	S
NM9 (Mir)	1816.07	515.64	327.42	32.97	≈ 412	S
NM6 (Cas)	607.06	567.34	772.31	12.33	≈ 421	S

TABLE III
OPERATING POINT SUMMARY OF CMFB DEVICES

Device	I_{DS} (μ A)	V_{GS} (mV)	V_{DS} (mV)	g_m (μ S)	V_{th} (mV)	Reg
PM14 (Ld)	-75.25	-557.96	-557.96	985.21	$\approx$ -455	S
PM13 (Ld)	-75.92	-557.96	-640.85	992.88	$\approx$ -455	S
NM21 (Sn)	37.62	584.79	921.67	722.71	$\approx$ 420	S
NM18 (Sn)	37.62	584.79	921.67	722.71	$\approx$ 420	S
NM20 (Mir)	33.74	579.62	838.77	658.91	$\approx$ 420	S
NM19 (Mir)	42.18	579.62	838.77	823.64	$\approx$ 420	S
NM17 (Tail)	151.16	515.64	320.38	2744.56	$\approx$ 412	S

VI. AI-ASSISTED DESIGN AND VERIFICATION

The following prompts were used during different stages of the design process, ranging from initial methodology selection to debugging and failure analysis.

Prompts Used

- I need to design a 5T OTA with a target g_m of 10 mS in 180 nm CMOS. Can you explain the g_m/I_D methodology and how I should use it to pick my initial bias current and transistor sizing? What overdrive voltage should I target for moderate inversion?
- I have a project coming up which is a more innovative version of the round 1 I had in Cadence. My task is to develop a 5T OTA with a 10 mS g_m which deviates by less than 1% using negative feedback. I am attaching the reference paper I went through. Could you break down how I should efficiently approach the problem, starting with attaining the DC operating point first?
- Here is my current replication/implementation[attached image] of the circuit. Currently I am not getting the required I_{DS} in the resistance follower part of the circuit. Can you check all the components' DC conditions and cross-verify with theoretical calculations to find out exactly which part of the circuit is not working as expected?
- I understood the part where we implement the resistance follower beta-multiplier circuit, but the feedback part of the circuit, could you break it down exactly how it is working and how it helps improve the overall result?

VII. FAILURE ANALYSIS AND SYSTEM LIMITATIONS

While the proposed architecture achieves a stable 10 mS transconductance under nominal conditions, several limitations define the boundaries of reliable operation. These arise primarily from device mismatch, voltage headroom constraints, and feedback loop behavior.

A. Sensitivity to Process and Geometric Mismatch

The design relies on a master-slave current replication mechanism, where the OTA transconductance is derived from a reference generated in the Beta-Multiplier. This approach assumes close matching between the bias network devices and the OTA input pair.

In practice, local process variations such as threshold voltage mismatch and lithographic inaccuracies introduce deviations in current density. Even small mismatches in device

dimensions can disrupt this symmetry, leading to a deviation in the replicated transconductance and reducing accuracy.

Additionally, the absolute value of g_m is set by the reference resistor:

$$g_m \approx \frac{1.5}{R_s}$$

Since on-chip resistors exhibit process-dependent variations (typically $\pm 10\%$ to 20%), the nominal transconductance may shift accordingly. While temperature stability is preserved, the absolute accuracy depends on resistor tolerance and layout matching techniques such as common-centroid placement and the use of dummy devices.

B. Voltage Headroom and Common-Mode Constraints

The use of stacked transistor structures in both the bias core and CMFB introduces constraints on voltage headroom.

In the OTA core, the tail current source operates with a limited saturation margin. A reduction in input common-mode voltage can reduce V_{DS} across the tail device, potentially pushing it into the triode region. This results in a loss of bias stability and reduced gain.

Similarly, a reduction in supply voltage limits the available headroom for the PMOS current mirrors in the bias network. If these devices leave saturation, the constant- g_m behavior degrades, and the bias reference becomes unreliable.

C. Feedback Loop Stability and Degenerate States

The system contains two feedback loops whose stability is critical to operation.

The Beta-Multiplier bias core includes a feedback mechanism that can exhibit a secondary equilibrium at zero current. Although the intended operating point is enforced under normal conditions, insufficient loop gain during startup or adverse conditions may cause the circuit to settle in this undesired state.

The CMFB loop regulates the output common-mode voltage by controlling the PMOS load devices. Its stability depends on the load capacitance at the output nodes. Variations in this load can shift the dominant pole of the loop, reducing phase margin and potentially leading to common-mode oscillations.

D. Summary

These limitations highlight that while the architecture achieves strong temperature stability, its practical performance depends on careful layout, adequate voltage headroom, and proper feedback loop design. Ensuring robust operation requires addressing mismatch, accounting for process variations, and maintaining sufficient phase margin in the feedback loops.